StressEar-AI: A Pre-Analysis Protocol for Public-Data Studies of Perceived Stress, Work-Related Factors, Tinnitus, and Hearing Outcomes

With a Cross-National External-Validation Plan and a Deterministic Referral-Safety Layer

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Abstract

Purpose: This is a *pre-analysis study protocol* for StressEar-AI (Stress–Ear Artificial Intelligence), not an empirical report. It prespecifies public-data analyses of the associations among *perceived stress and work-related factors*, tinnitus, and hearing outcomes in adults, together with a cross-national external-validation plan and a clinician-governed, safety-gated artificial-intelligence (AI) architecture. We deliberately avoid causal overclaiming: because the population surveys do not contain validated occupational-stress instruments, the exposure is labeled perceived stress and work-related factors—not measured occupational stress—and is treated as potentially *associated* with tinnitus burden and hearing-related outcomes rather than as a proven cause of cochlear injury. As a proof of concept, we additionally report *preliminary real-data results* from NHANES; the primary perceived-stress analysis is reserved for KNHANES.

Method (prespecified): The Korea National Health and Nutrition Examination Survey (KNHANES) is the development dataset and the U.S. National Health and Nutrition Examination Survey (NHANES) is the geographic external-validation dataset; we fix the survey cycles, age windows, audiometric frequencies, exclusions, and expected analytic sample sizes in advance. A causal directed acyclic graph (DAG) defines a minimal sufficient adjustment set and separates confounders from mediators (e.g., depressed mood, sleep), with primary analyses avoiding overadjustment and full-adult versus employed-restricted analyses addressing the healthy-worker effect. Endpoints (tinnitus presence, bothersome tinnitus, better- and worse-ear pure-tone average, and asymmetry) are modeled with survey-weighted regression; two *research-only* risk-stratification models are evaluated with discrimination, calibration-in-the-large and calibration slope (before and after recalibration), decision-curve utility, common-support checks, and subgroup fairness (including socioeconomic status and race/ethnicity), reported per STROBE and TRIPOD+AI. A deterministic red-flag layer for sudden sensorineural hearing loss (SSNHL) overrides model output; its evaluation prespecifies vignette composition, dual otolaryngology/audiology labeling, inter-rater agreement, and sensitivity with confidence intervals. An open medical language model (e.g., MedGemma) is confined to schema-constrained extraction verified by clinicians.

Contributions: (1) a survey-weighted public-data framework that distinguishes stress–tinnitus from stress–audiometric-threshold associations without unsupported causal claims; (2) a prespecified KNHANES→NHANES external-validation plan reporting calibration, recalibration, clinical utility, and subgroup fairness; (3) a clinically governed AI architecture in which schema-constrained extraction is clinician-verified and a deterministic red-flag layer overrides probabilistic predictions for time-critical hearing symptoms; and (4) an open, reproducible scaffold of harmonized variable mappings, executable code, model cards, safety tests, and a synthetic-data pathway.

Preliminary results (NHANES, adults 40–69): Survey-weighted prevalence was 28.5% (95% CI 26.5–30.4) for tinnitus and 12.0% (10.4–13.6) for better-ear hearing loss (>25 dB HL). In mutually adjusted design-based models, tinnitus was associated with occupational noise (OR 1.59, 1.32–1.92) and a depression-screen proxy (OR 1.61, 1.23–2.11), whereas hearing loss was dominated by age (OR 1.11/year) and the depression proxy (OR 2.29, 1.56–3.35) with only a weak, non-significant noise association—a pattern consistent with the framework's hypotheses. These are associations, not causal effects.

Conclusions: Registered public-data analyses can rigorously characterize associations and benchmark explainable, safety-gated AI, but prospective clinical cohorts remain necessary for treatment timing, recovery, and prognosis. StressEar-AI fixes the design, safety governance, and reproducible scaffold—now demonstrated end-to-end on

real NHANES data—ahead of the primary KNHANES analysis.

Keywords: tinnitus; bothersome tinnitus; hearing loss; audiometry; perceived stress; work-related factors; sudden sensorineural hearing loss; KNHANES; NHANES; external validation; explainable AI; open medical language models; TRIPOD+AI; study protocol; reproducibility; audiology.

1 Introduction

Tinnitus and hearing loss are among the most common auditory health concerns worldwide, affecting communication, sleep, emotional well-being, work participation, and quality of life. The World Health Organization estimates that more than 1.5 billion people live with some degree of hearing loss, a number projected to rise substantially by 2050 (World Health Organization, 2021). Tinnitus alone affects an estimated 10–15% of adults, and a clinically meaningful subset experiences bothersome tinnitus that interferes with concentration and mood (Tunkel et al., 2014; McCormack et al., 2016). In employed middle-aged adults, tinnitus frequently emerges in a context of psychosocial stress, long working hours, sleep disruption, occupational or recreational noise, cardiometabolic risk, and delayed help-seeking. Because these factors are temporally and biologically intertwined, simple causal narratives are unreliable and can mislead both patients and clinicians.

1.1 Motivating clinical observation

The framework is motivated by a common and instructive clinical trajectory, presented here as an anonymized composite rather than as identifiable patient data. A worker in the late 40s to early 50s first experiences unilateral tinnitus in one ear during a period of heavy workload. Reasoning intuitively, the person "rests" the affected ear by shifting phone calls and earphone use to the contralateral side; over roughly nine months the tinnitus resolves. About a year later, the previously unaffected ear develops sudden aural fullness and a "pop" during a stressful workday, followed within days to weeks by reduced audibility of soft sounds. Lacking awareness of the narrow treatment window for sudden sensorineural hearing loss (SSNHL), the person delays evaluation for roughly two months; by the time intratympanic and systemic steroid therapy is attempted, recovery is incomplete. This trajectory highlights two clinically actionable points that StressEar-AI (Stress–Ear Artificial Intelligence) is built around: (i) stress-associated tinnitus is common and often self-limiting, but (ii) sudden or rapidly progressive hearing loss is a medical urgency in which delayed care—not stress per se—may drive irreversible impairment. Attributing acute hearing symptoms to "stress" without audiological and otolaryngological assessment is therefore hazardous.

1.2 Conservative premise and exposure naming

StressEar-AI begins from a conservative and testable premise: *perceived stress and work-related factors* may be *associated* with tinnitus and hearing outcomes, but public cross-sectional datasets are primarily suited to estimating associations, not to proving that stress directly causes tinnitus or sensorineural hearing loss. We adopt a strict naming principle throughout: because KNHANES and NHANES contain general perceived-stress, mood, and employment items rather than validated job-strain instruments (e.g., the Job Content Questionnaire or Effort–Reward Imbalance scale), we never describe the exposure as measured "occupational stress." It is labeled *perceived stress* (the psychological-distress/stress item) with *work-related factors* (occupation, employment status, and, where available, working hours and shift work) modeled as distinct covariates or effect modifiers. This distinction is central to the framework's statistical claims and to its clinical safety posture.

1.3 This is a protocol, not a results paper

This manuscript is a pre-analysis protocol. It contains *no* epidemiological findings; the only numbers reported are synthetic pipeline checks (clearly labeled as such) that verify the code computes and reports metrics correctly. Fixing the design, variable definitions, survey cycles, adjustment sets, and evaluation plan *before* any real-data analysis is a deliberate guard against analyst degrees of freedom (e.g., selecting cycles or covariate sets after seeing results). Reviewers and readers should therefore judge the contribution on the rigor, safety governance, and reproducibility of the plan, not on empirical results, which are reserved for a subsequent report.

1.4 Aims and hypotheses

The framework has four aims, each with prespecified, falsifiable hypotheses:

1. **Association (development).** Estimate weighted associations between perceived stress and (a) tinnitus status and (b) audiometric hearing loss in KNHANES, adjusting for noise exposure and health covariates. *H1: A positive association between perceived stress and tinnitus persists after adjustment; the association with audiometric thresholds is weaker and may attenuate to non-significance after adjusting for noise and age.*

2. **Transportability (external validation).** Test whether harmonized feature definitions and prediction models developed in KNHANES retain discrimination and calibration in NHANES. *H2: Discrimination degrades modestly but calibration can be restored by recalibration, supporting cross-national transportability of the tinnitus model more than the hearing-loss model.*

3. **Explainable, safe AI.** Evaluate whether an open medical language model can extract audiology-relevant fields from deidentified narratives at clinician-verified accuracy while a deterministic red-flag layer reliably routes SSNHL and neurologic warning signs to urgent referral. *H3: Schema-constrained extraction reaches high human-verified agreement with a low unsupported-output rate, and the safety layer achieves near-complete red-flag recall.*

4. **Open, shared benefit.** Provide an openly licensed, reproducible pipeline (code, data dictionary, model cards, prompts) that lowers the barrier for researchers to reproduce analyses and add local cohorts.

1.5 Contributions

This protocol makes four contributions. *First*, it establishes a survey-weighted, public-data framework that distinguishes stress–tinnitus associations from stress–audiometric-threshold associations without unsupported causal claims. *Second*, it prespecifies cross-national external validation from KNHANES to NHANES, including calibration, recalibration, clinical utility, and subgroup fairness. *Third*, it introduces a clinically governed AI architecture in which schema-constrained language-model extraction is clinician-verified and a deterministic red-flag layer overrides probabilistic predictions for time-critical hearing symptoms. *Fourth*, it releases an open, reproducible scaffold comprising harmonized variable mappings, executable code, model cards, safety tests, and a synthetic-data pathway.

1.6 Scope and positioning

To keep the message focused, this AJA-targeted protocol concentrates on *three* axes: (i) KNHANES–NHANES associations among perceived stress/work-related factors, tinnitus, and hearing; (ii) cross-national external validation of *research-only* risk-stratification models; and (iii) a safe referral layer for sudden hearing loss. Any prediction model here is explicitly a research risk-stratification tool—not a screening, diagnostic, or triage device—with intended use specified in Table 5. Optional neurophysiological substudies (OpenNeuro tinnitus imaging/EEG) and audiogram-digitization of legacy records are mentioned only briefly as future directions and are out of scope for the primary analyses. AI is an enabling research tool throughout, never a replacement for clinical judgment. As a pre-analysis protocol, the manuscript reports the design, safety governance, and reproducible scaffold; it reports no epidemiological findings, which await the analyses specified herein.

2 Related Work

2.1 Epidemiology of tinnitus, hearing loss, and stress

Population studies have repeatedly shown that tinnitus and hearing loss often coexist, but their relationships are heterogeneous. Analyses using KNHANES have estimated tinnitus prevalence and associated factors in Korea and demonstrated that nationally representative data can support clinically meaningful tinnitus research (Park et al., 2014; Joo et al., 2015). NHANES has similarly enabled U.S. studies of audiometry, occupational noise exposure, tinnitus, and mental-health correlates (Hoffman et al., 2020; Chakrabarty et al., 2024; Mahboubi et al., 2013). Reviews of tinnitus epidemiology emphasize wide prevalence estimates driven by heterogeneous case definitions, underscoring the need for harmonized variables when pooling or comparing across surveys (McCormack et al., 2016). Evidence linking psychological stress to tinnitus onset and severity is consistent but largely observational; longitudinal occupational cohorts suggest stress is associated with incident tinnitus, yet residual confounding by noise, sleep, and cardiometabolic factors is difficult to exclude (Baigi et al., 2011).

2.2 Beyond the pure-tone average

Recent audiology work highlights the need to measure more than the better-ear pure-tone average. Studies of executive control and attention in individuals with tinnitus suggest that cognitive burden may be relevant to perceived functional impact, and speech-perception-in-noise and hearing-aid signal-management research shows that real-world listening difficulty can persist even when threshold-based metrics appear near-normal. These directions motivate StressEar-AI to separate tinnitus presence from bothersome tinnitus and to analyze worse-ear and per-ear thresholds and asymmetry, not better-ear PTA alone. (Specific recent *AJA* exemplars will be cited with verified bibliographic details in the submission version; placeholder entries have been removed from this draft to avoid unverifiable references.)

2.3 Clinical pathways and the treatment window

Clinical practice guidelines frame the endpoints StressEar-AI must respect. The clinical practice guideline for tinnitus distinguishes bothersome from non-bothersome tinnitus and prioritizes education, cognitive-behavioral strategies, and sound therapy over pharmacologic cures (Tunkel et al., 2014). Critically, the guideline for sudden hearing loss defines SSNHL as a medical urgency: prompt audiometric confirmation and early corticosteroid therapy within roughly two weeks offer the best chance of recovery, and delayed presentation is associated with worse outcomes (Chandrasekhar et al., 2019). This "golden-window" principle is the clinical rationale for the deterministic red-flag layer described in Section 5.

Prior work involving Hun Yi Park and colleagues is especially relevant for the clinical expansion of this study. Sound-localization findings in unilateral hearing-aid users show that the side of hearing impairment may influence spatial-hearing interpretation (Ha et al., 2022). Early-management research in traumatic benign paroxysmal positional vertigo, although not a tinnitus study, underscores the importance of rapid identification and pathway-based care for acute otologic and vestibular symptoms (Kim et al., 2022). Cochlear-implant research using a CT-derived basal turn–facial ridge angle illustrates how explainable anatomical variables can support prediction of residual hearing preservation (Kim et al., 2021). Together, these studies support a StressEar-AI design that separates screening, referral, treatment, and rehabilitation endpoints.

2.4 Open medical AI and reporting standards

Open medical AI models are increasingly available for research use. MedGemma is a collection of open models optimized for medical text and image comprehension, positioned as a starting point for health-AI development rather than a validated diagnostic device (Google Research, 2025; Google Health AI, 2026). For StressEar-AI, this implies constraining the model to schema extraction and evidence-linked explanations, with clinician verification and external validation before any clinical use. Methodologically, the framework adheres to established reporting standards for observational and prediction research—STROBE for observational analyses (von Elm et al., 2007) and the TRIPOD+AI statement for AI-based prediction models (Collins et al., 2024)—so that reported models are transparent, comparable, and auditable.

3 Public Dataset Selection

Dataset selection was governed by four requirements for the primary association analyses: (i) survey-weighted population representativeness; (ii) objective pure-tone audiometry; (iii) tinnitus and/or hearing-difficulty items; and (iv) covariates for noise exposure, psychosocial stress, and general health. Only KNHANES and NHANES jointly satisfy all four at population scale; other public resources are retained as optional secondary datasets. Table 2 summarizes the selection, and Table 3 maps target variables to each source.

3.1 Primary Development Dataset: KNHANES

KNHANES is selected as the primary development dataset because it is nationally representative for Korea and has previously supported tinnitus and hearing-loss analyses (Park et al., 2014; Joo et al., 2015). Its advantages include Korean population relevance, complex-survey weights, pure-tone audiometry in applicable cycles (otologic examination cycles from 2009–2012 onward), tinnitus and hearing-difficulty questionnaire items in selected cycles, health and behavioral covariates, and psychosocial or stress-related self-report variables (e.g., perceived-stress and depressed-mood items). It also provides a realistic bridge to future Korean hospital cohorts, including the planned Ajou University Hospital extension. Because item wording and audiometry availability differ across cycles, the analysis prespecifies a per-cycle codebook review and restricts pooling to cycles with comparable definitions.

The limitations are equally important. KNHANES is not a workplace cohort, and the audiometric analyses are cross-sectional. It can therefore evaluate *associations* among perceived stress, work-related variables, tinnitus, and audiometric outcomes, but cannot by itself establish that occupational stress *caused* tinnitus or hearing loss. Treatment timing and recovery after sudden hearing loss are not captured.

3.2 External-Validation Dataset: NHANES

NHANES is selected as the primary external-validation dataset because it provides openly downloadable U.S. public-use files, pure-tone audiometry and tympanometry in selected cycles (e.g., 2011–2012, 2015–2016, 2017–March 2020), occupational and recreational noise-exposure items, tinnitus and hearing questionnaires in selected cycles, and extensive demographic and health covariates. NHANES enables testing whether feature definitions and risk models derived from KNHANES retain discrimination and calibration across population, language, health-system, and occupational contexts. As with KNHANES, complex-survey weights, strata, and primary sampling units must be honored, and cycle-specific age eligibility for the audiometry module must be respected.

3.3 Optional Secondary Datasets

Open audiology repositories may be used for targeted secondary experiments: the Oldenburg Hearing Health Repository and Zenodo auditory-test datasets for audiometric model benchmarking and audiogram-based clustering; OpenNeuro tinnitus neuroimaging and fNIRS datasets for an optional neurophysiological substudy of tinnitus subtypes; and Mendeley/Zenodo tinnitus EEG datasets for explainable-biomarker exploration. These resources are not primary because they typically lack the joint combination of population weights, tinnitus items, audiometry, occupational factors, and general-health covariates required for the main survey-weighted analyses. Any such dataset is used only after verifying its license, consent basis, and permission for secondary research use.

3.4 Prospective Clinical Extension (Ajou University Hospital)

A prospective otolaryngology/audiology cohort is specified to address what public cross-sectional data cannot: treatment timing, SSNHL recovery trajectories, tinnitus burden over time, and hearing-aid rehabilitation. This cohort contributes deidentified audiograms and clinical narratives for clinical external validation and provides the substrate for the AI extraction and safety evaluations. Raw clinical data are not redistributed; only derived, deidentified, or synthetic artifacts are shared through the open repository, subject to institutional review board (IRB) approval and data-governance review.

4 Methods (Prespecified)

This pre-analysis protocol is reported following STROBE for the observational analyses (von Elm et al., 2007) and TRIPOD+AI for the prediction models (Collins et al., 2024); the prospective clinical extension will follow SPIRIT. A completed reporting checklist accompanies the repository. All design choices below are fixed prior to any real-data analysis.

4.1 Study Design

The work has three stages. **Stage 1** analyzes KNHANES as the primary public-data development cohort (association estimation and research-model development). **Stage 2** evaluates cross-national transportability in NHANES (geographic external validation). **Stage 3** prepares a prospective clinical extension at Ajou University Hospital for clinical external validation, treatment-timing analysis in SSNHL, and evaluation of the AI extraction and safety components. Stages 1–2 use only deidentified public-use files; Stage 3 proceeds only under IRB approval.

4.2 Fixed Datasets, Cycles, and Eligibility

To eliminate analyst degrees of freedom, the survey cycles, files, ages, frequencies, and exclusions are fixed in advance (Table 2). **KNHANES:** the otologic-examination cycles carrying pure-tone audiometry and tinnitus items (KNHANES IV–V, 2009–2012) form the primary development window; cycles are pooled only after confirming identical item wording and audiometry protocols. **NHANES:** the audiometry-module cycles 2011–2012, 2015–2016, and 2017–March 2020 form the external-validation window. For both, eligible records are adults aged 40–69 years (the band jointly covered by the audiometry modules and our target of mid-to-late-career workers), with valid examination weights and a non-missing value for the outcome under analysis. Pure-tone thresholds are analyzed at 0.5, 1, 2, 3, 4, 6, and 8 kHz per ear; conductive patterns are flagged and, where tympanometry/air–bone data permit, excluded in a sensitivity analysis. We prespecify expected raw and analytic sample sizes per window (reported in the repository data dictionary) and the exact set of predictors available in *both* surveys; any predictor available in only one survey is excluded from the common transportable model and used only in country-specific extended models.

4.3 Exposure: Perceived Stress and Work-Related Factors

Consistent with the naming principle (Introduction), the primary exposure is *perceived stress*, operationalized from each survey's general stress/psychological-distress item, and *work-related factors* (occupation category, employment status, and—where a cycle provides them—working hours and shift work) are modeled as separate covariates or effect modifiers, never as a proxy for validated job strain. Because item wording, response categories, and cultural response styles differ across surveys, Table 4 records, for every exposure and covariate, the original question, response scale, harmonization rule, and a measurement-comparability rating (high/medium/low). Variables rated *low* comparability (typically the stress and some psychosocial items) are (a) excluded from the common transportable model, (b) entered only in country-specific models, and (c) examined in a formal measurement-invariance check (configural/metric where a latent construct is defined). We do not pool a variable across countries unless at least metric-level comparability is plausible.

4.4 Causal Structure and Adjustment Sets

Figure 2 presents the directed acyclic graph (DAG) encoding our assumptions. Age, sex, education, occupation/employment, and noise exposure are treated as confounders of the perceived-stress→outcome relationships and constitute the **minimal sufficient adjustment set** for the primary analysis. Depressed mood and sleep are treated as *potential mediators* of the stress→tinnitus path (and possibly colliders when conditioning opens non-causal paths); conditioning on them can constitute overadjustment. Accordingly:

- **Primary model** adjusts for the minimal sufficient set only (age, sex, education, occupation/employment, noise).
- **Extended model** additionally includes depressed mood and sleep, reported separately and interpreted as a mediator-adjusted (not confounder-adjusted) estimate.
- **Mediation analysis** decomposes total, direct, and indirect (through depressed mood/sleep) associations under stated assumptions, as an exploratory addendum.
- **Collider check** enumerates conditioning sets that could induce collider bias and excludes them from the primary specification.

Because restricting to employed participants can induce a healthy-worker effect (and a selection collider), the primary population is *all eligible adults*, with an *employed-restricted* analysis reported as a prespecified secondary contrast rather than the main result.

4.5 Outcomes

Endpoints are defined a priori and separated:

- **Tinnitus presence** (any tinnitus in the past 12 months) and **bothersome tinnitus** (tinnitus causing annoyance/sleep or concentration difficulty) are modeled as *distinct* endpoints, since their determinants differ.
- **Hearing loss** is quantified by the speech-frequency pure-tone average ($\mathrm{PTA}_{0.5,1,2,4}$) for *both the better ear and the worse ear*, plus a **per-ear** (ear-level, mixed-effects) analysis, because a better-ear summary can mask the unilateral/asymmetric presentations most relevant to SSNHL. The primary threshold is PTA $>$ 25 dB HL (the WHO grade-1 cutoff (World Health Organization, 2021)); $>$ 20 dB HL and high-frequency $\mathrm{PTA}_{3,4,6}$ are prespecified sensitivity definitions, with the 25 dB HL choice justified as the established clinical cutpoint.
- **Asymmetric hearing loss** is defined as an interaural difference $\geq$15 dB at $\geq$2 adjacent frequencies, consistent with asymmetry criteria used in sudden/asymmetric hearing-loss guidance (Chandrasekhar et al., 2019).
- **Secondary:** self-reported hearing difficulty, hearing-aid use, and quality-of-life correlates.

Noise exposure is coded, where cycles permit, not merely as a binary but with duration, intensity/source (occupational, recreational, firearm), and hearing-protection use, so residual confounding by noise is minimized.

4.6 Statistical Analysis

All public-data analyses use survey-aware methods (Taylor-linearization variance estimation honoring weights, strata, and PSUs). Descriptive analyses estimate weighted prevalence with 95% confidence intervals. Association analyses use weighted logistic, ordinal, or linear regression by outcome type. The **primary specification adjusts for the minimal sufficient set**; the extended (mediator-adjusted) model is secondary. Results are reported as associations (odds ratios or mean differences) with 95% CIs, explicitly not as causal effects. Multiplicity is handled by prespecifying primary hypotheses (Introduction) and controlling the false-discovery rate across secondary contrasts.

Missing data. Missingness is described by variable and mechanism. Primary analyses use multiple imputation by chained equations under a missing-at-random assumption (design variables in the imputation model), pooled by Rubin's rules; complete-case analysis is a sensitivity check. In external validation, records missing a common predictor are handled by the same prespecified rule in both surveys.

Sample size and precision. Sample sizes are fixed, so we frame precision rather than post-hoc power: with several thousand audiometry participants per window, weighted prevalence is estimable to within a few percentage points and moderate associations (OR $\approx$ 1.3–1.5) are detectable at conventional error rates. Exact expected analytic N per window is fixed in the repository before analysis.

4.7 Research-Only Prediction Models and Intended Use

Two targets—tinnitus presence and audiometric hearing loss—are developed as *research risk-stratification models*, not screening, diagnostic, or triage devices. Their intended use, target user, prediction time, inputs, decision, and the costs of false positives/negatives and contraindicated uses are specified in Table 5. We first fit transparent survey-weighted logistic baselines with prespecified predictors, then gradient-boosted trees as a comparator; the simpler model is preferred unless the complex model yields a clinically meaningful, calibrated improvement. Development uses KNHANES with repeated stratified cross-validation and optimism correction. All targets, predictors, and metrics are frozen before external validation (Table 7). SHAP values are reported for interpretability, with the explicit caveat that SHAP does not guarantee causal or even stable attributions when predictors are correlated; importance is interpreted descriptively, not causally.

4.8 External Validation Across Survey Designs

Transporting a KNHANES model to NHANES confronts differences in weighting, population prevalence, age range, category coding, and health-system context. We therefore prespecify:

- **Unweighted vs. weighted performance** reported separately—unweighted for individual-level prediction, survey-weighted for population-level performance.
- **Calibration-in-the-large and calibration slope reported separately**, before and after recalibration (intercept-only, then intercept+slope update of the frozen model).
- **Common-support analysis** restricting comparison to the overlapping covariate region of the two surveys.
- **Case-mix standardization** to distinguish genuine model transportability from case-mix differences in prevalence/age.
- **A reduced common model** (predictors available and comparable in both surveys) versus **country-specific extended models**, so that limited measurement invariance does not masquerade as poor transportability.

Fairness and subgroup evaluation. Discrimination and calibration are reported within prespecified subgroups: sex, age band, employment status, noise-exposure status, unilateral/asymmetric hearing, *socioeconomic status* (education/income), and *race/ethnicity* (NHANES, where collected). Subgroup calibration and error gaps are first-class results, not afterthoughts.

4.9 Sensitivity Analyses

Prespecified sensitivity analyses test alternative hearing-loss definitions (25 vs. 20 dB HL; high-frequency PTA; worse-ear and per-ear), age restrictions, exclusion of conductive patterns when tympanometry permits, alternative stress operationalizations, mediator- vs.

confounder-treatment of depressed mood/sleep, all-adult vs. employed-restricted populations, and stratification by noise exposure, sex, and asymmetry.

4.10 Ethics and Governance

Stages 1–2 use publicly available, deidentified survey data and do not constitute human-subjects research requiring new consent; KNHANES and NHANES were approved by their respective review boards, and their use follows each provider's data-use terms. The Stage-3 clinical extension will obtain IRB approval at Ajou University Hospital, with a data-management plan covering deidentification, access control, and secondary-use consent. No identifiable patient data are placed in the public repository; only derived, deidentified, or synthetic artifacts are shared.

5 Open AI System and Research Acceleration

5.1 Motivation

A major barrier to audiology AI is the scarcity of harmonized, reusable clinical features. StressEar-AI addresses this by publishing a transparent feature schema, reproducible public-data code, and a clinician-verification protocol. This supports a broadly shared-benefit research model: investigators can reuse the pipeline, add local cohorts, and compare models without exposing identifiable patient data.

5.2 Model Roles and Prohibitions

The AI system has four restricted roles: (1) it harmonizes variable names and labels across public datasets; (2) it trains transparent baseline models for association and calibrated risk stratification; (3) in the clinical extension, an open medical language model (e.g., MedGemma) extracts structured fields from deidentified notes using a fixed JSON schema; and (4) it drafts explanations that cite only approved study documents or clinical guidelines. The system is explicitly prohibited from diagnosing an individual patient, assigning a definitive etiology, recommending or dosing steroids, or delaying urgent referral. These prohibitions are enforced in code and documented in a model card.

5.3 Deterministic Red-Flag Safety Layer

Before any probabilistic model output is surfaced, a deterministic rule layer screens for red flags derived from the SSNHL and tinnitus guidelines (Chandrasekhar et al., 2019; Tunkel et al., 2014): sudden or rapidly progressive hearing loss (especially unilateral), aural fullness with a discrete onset, single-sided tinnitus with asymmetry, vertigo, or focal neurologic signs. Any trigger overrides the model and returns an "urgent audiology/otolaryngology referral" message. This operationalizes the "golden-window" lesson from the motivating vignette: the framework must never let a probabilistic "low-risk, likely stress" estimate suppress timely referral.

Evaluation design. The safety layer's evaluation is prespecified in detail (Table 8) rather than asserted as a target recall alone. A vignette set of $\geq$200 cases mixes real deidentified, expert-authored, and synthetic cases in a fixed ratio; each is labeled by at least two otolaryngology/audiology experts with inter-rater agreement (κ) reported before adjudication. We report sensitivity (red-flag recall) with 95% CIs as the primary metric and specificity/over-referral as secondary, stratified by laterality, onset speed, and vertigo/neurologic signs, with explicit robustness tests for negation, ambiguous temporal expressions, and colloquial patient phrasing. Every missed red flag is audited as a critical failure. Crucially, we state that a sensitivity of 1.0 on a finite test set is a target, *not* a guarantee of zero misses in deployment; the layer is a safeguard layered on top of, never a replacement for, clinical assessment.

5.4 Schema-Constrained Extraction and Governance

LLM extraction is constrained to a fixed schema with typed fields and controlled vocabularies; free-text generation is disallowed for structured outputs. Each extracted field carries a source span and a confidence flag, and every field is clinician-verified before entering analysis. Prompts, schema versions, model weights/version, temperature, and random seeds are logged. We define the *confidence flag* explicitly as a rule-based reliability indicator (presence/agreement of a supporting source span), not the language model's own token probability, to avoid conflating self-reported confidence with correctness.

Extraction evaluation. Because "field accuracy" alone is inadequate for structured extraction, we report per field: exact-match rate; precision, recall, and F1; omission rate; evidence-span agreement; and dedicated accuracy for temporal expressions, laterality, onset speed, and symptom duration. We further report a clinically significant error rate (errors that could change referral), document-level exact-match rate, run-to-run consistency across repeated generations, clinician correction time, and performance by model version. Targets and thresholds are fixed before evaluation.

5.5 Robustness and Trustworthiness

Trustworthiness is pursued through eight design choices: (1) public-data baselines; (2) country-level external validation; (3) survey-weighted epidemiology; (4) interpretable models before complex models; (5) locked feature and outcome definitions; (6) clinician review of LLM-extracted variables; (7) subgroup calibration and error analysis; and (8) full run provenance (dataset versions, variable mappings, hyperparameters, prompts, seeds). Reproducibility is supported by pinned dependencies and deterministic seeds.

5.6 Open-Science Contribution

The repository includes executable analysis templates, a data dictionary, model cards, prompt templates, a red-flag test suite, and a completed reporting checklist. Because KNHANES files cannot be redistributed without official access procedures, the code specifies directory expectations and variable-mapping scripts rather than repackaging raw data. NHANES download helpers retrieve public files directly from CDC endpoints where available. A synthetic-data generator lets others exercise the full pipeline end-to-end without any restricted data, lowering the barrier to reuse and review.

6 Results

This section reports *preliminary real-data results* from NHANES (a demonstration of the framework on genuine public-use data), followed by the analyses that remain planned for the primary KNHANES development cohort. The NHANES analysis is illustrative rather than the primary test of the aims: NHANES 2011–2018 lacks a validated perceived-stress instrument, so the available psychological-distress proxy is the PHQ-9 depression screen, which the study DAG treats as a *mediator*. The primary perceived-stress exposure analysis is therefore reserved for KNHANES, which carries a general stress item.

6.1 Planned primary analyses (KNHANES; not yet reported)

Per the prespecified aims, the KNHANES analyses will report: (i) weighted prevalence of tinnitus presence, bothersome tinnitus, and hearing impairment with 95% CIs; (ii) primary (minimal-adjustment-set) associations between perceived stress and each endpoint, with the extended mediator-adjusted model separate; (iii) associations with better-, worse-ear, and per-ear thresholds, expected to attenuate after adjusting for age and noise (H1); (iv) all-adult and, secondarily, employed-restricted analyses (healthy-worker check); (v) KNHANES→NHANES external validation with unweighted/weighted performance, calibration-in-the-large and slope before/after recalibration, common support, and case-mix standardization (H2); and (vi) subgroup fairness including SES and race/ethnicity.

6.2 Reproducible scaffold demonstration (synthetic; not a result)

To show that the pipeline runs end-to-end and produces correctly structured, TRIPOD+AI-aligned outputs, the repository ships a synthetic-data demonstration. **These numbers are artifacts of randomly generated data and carry no epidemiological meaning**; they exist only to verify wiring, metric computation, and reporting format, and are deliberately kept out of the abstract and any headline. On the synthetic sample the tinnitus target yields AUROC ≈ 0.64, AUPRC ≈ 0.49, Brier ≈ 0.20, and the hearing-loss target yields an analogous record (Table 7). Substituting real KNHANES/NHANES files following the documented layout routes the identical code path to the design-consistent estimates specified above.

6.3 Preliminary real-data results (NHANES public-use)

Using the tested, design-based pipeline (`code/src/nhanes_analysis.py`; estimators unit-tested in `code/src/test_nhanes_analysis.py`), we analyzed real NHANES public-use files for U.S. adults aged 40–69, pooling the 2011–2012, 2015–2016, and 2017–2018 cycles (Table 1).

Prevalence. Survey-weighted prevalence was 28.5% (95% CI 26.5–30.4; n = 5,354) for any tinnitus in the past 12 months, 12.0% (10.4–13.6; n = 4,808) for better-ear speech-frequency hearing loss (>25 dB HL), and 8.4% (7.5–9.4; n = 7,293) for a positive PHQ-9 depression screen.

Associations. In mutually adjusted design-based logistic models (adjusting for occupational noise, depressed mood, age, and sex), *tinnitus* was associated with occupational-noise exposure (OR 1.59, 1.32–1.92), depressed mood (OR 1.61, 1.23–2.11), and female sex (OR 1.61, 1.35–

Table 1: **Real NHANES results** (U.S. adults 40–69; pooled 2011–2012, 2015–2016, 2017–2018). Prevalence is design-based (survey-weighted, Taylor-linearized); associations are odds ratios (OR) from design-weighted logistic regression, mutually adjusted, and are *not* causal. Depressed mood is a PHQ-9 screen used here as the available psychological-distress proxy (NHANES lacks a validated perceived-stress instrument); per the study DAG it is a mediator, so its OR is a mediator-level association. Generated by `code/src/nhanes_analysis.py`. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

	Tinnitus	Hearing loss (>25 dB HL)
Prevalence (95% CI)	28.5% (26.5–30.4)	12.0% (10.4–13.6)
Adjusted associations, OR (95% CI)		
Occupational noise	1.59 (1.32–1.92)***	1.22 (0.94–1.58)
Depressed mood (PHQ-9≥10)	1.61 (1.23–2.11)***	2.29 (1.56–3.35)***
Age (per year)	1.00 (0.98–1.01)	1.11 (1.09–1.13)***
Female sex	1.61 (1.35–1.91)***	0.42 (0.29–0.61)***
Analytic N (associations)	4603	4463
Depression (PHQ-9≥10) prevalence: 8.4% (7.5–9.4), $N = 7293$		

1.91), but not with age (OR 1.00, 0.98–1.01). *Hearing loss* was associated with age (OR 1.11 per year, 1.09–1.13) and depressed mood (OR 2.29, 1.56–3.35), inversely with female sex (OR 0.42, 0.29–0.61), and only weakly and non-significantly with occupational noise (OR 1.22, 0.94–1.58). These are associations, not causal effects.

Interpretation (a priori pattern). The pattern is coherent with the framework's hypotheses. The psychological-distress proxy is associated with both outcomes, but more than doubles the odds of *audiometric* hearing loss while showing a smaller association with *tinnitus*; age dominates hearing loss but is unrelated to tinnitus in this band; and the sex pattern (higher tinnitus, lower hearing loss in women) matches known epidemiology. Because the proxy is a mediator in the DAG, these are mediator-level associations, and NHANES lacks the perceived-stress exposure; they therefore motivate—but do not substitute for—the primary KNHANES stress analysis. All values are reproduced by the committed results file and can be regenerated end-to-end from the CDC files.

6.4 Planned AI-component reporting

LLM-extracted fields will be reported with the per-field metric suite of Section 5 (exact match; precision/recall/F1; omission; span agreement; temporal/laterality/onset/duration accuracy; document-level match; run consistency; clinically significant error rate), plus clinician correction time and per-version performance. The red-flag layer will be reported per Table 8 (dual-expert-labeled vignettes; sensitivity with 95% CI; specificity/over-referral; subgroup and robustness), with any missed red flag audited as a critical failure and with the explicit caveat that finite-set perfection does not guarantee zero deployment misses. Model cards accompany each released model, documenting intended use (Table 5), data provenance, subgroup metrics, and prohibitions.

7 Discussion

As a protocol, StressEar-AI makes three focused commitments along the axes defined in the Introduction. First, it selects KNHANES for development and NHANES for cross-national external validation, and prespecifies—rather than post-hoc selects—cycles, variables, adjustment sets, and metrics. Second, it reframes AI as a reproducible research accelerator bounded by a deterministic safety layer, not a diagnostic authority. Third, it operationalizes reproducibility through open code, a data dictionary, model cards, safety tests, and a synthetic-data path.

7.1 Anticipated interpretation (a priori)

The expected scientific contribution is *not* a claim that workplace stress damages hearing. Rather, the planned analyses ask whether *perceived stress* is consistently associated with tinnitus and audiometric outcomes after the minimal sufficient adjustment. We hypothesize a stress–tinnitus association that is more robust than the stress–threshold association, the latter expected to attenuate after adjusting for age and noise (H1). If confirmed, this asymmetry has a clear reading: perceived stress may relate to tinnitus perception and burden—a partly central, attention-modulated phenomenon—more than to cochlear thresholds. Because depressed mood and sleep are modeled as mediators, the primary estimate avoids overadjusting away a genuine indirect pathway; the mediation analysis quantifies it explicitly.

7.2 Preliminary NHANES findings

The real-data demonstration is encouraging and internally coherent. The psychological-distress proxy (a positive PHQ-9 screen) was associated with both tinnitus (OR 1.61) and, more strongly, audiometric hearing loss (OR 2.29), while occupational noise was associated with tinnitus (OR 1.59) but not significantly with better-ear hearing loss (OR 1.22, $p = 0.14$), and age dominated hearing loss (OR 1.11 per year) but was unrelated to tinnitus. This dissociation—distress and noise relating to tinnitus, while age and distress relate to audiometric thresholds—mirrors the asymmetry the framework anticipated between symptom-level and threshold-level outcomes, and the sex pattern (higher tinnitus, lower hearing loss in women) is consistent with prior epidemiology. Three caveats bound these results: they are cross-sectional associations, not causal effects; the exposure is a depression *proxy* that the DAG treats as a mediator, not the perceived-stress exposure itself; and better-ear PTA understates unilateral disease, so the planned worse-ear and per-ear analyses may differ. They nonetheless show the pipeline yields sensible, reproducible estimates on genuine public data and motivate the primary KNHANES stress analysis.

7.3 Clinical contribution and the treatment window

The most consequential stance is procedural rather than statistical: because SSNHL is a time-critical emergency (Chandrasekhar et al., 2019), the system is engineered so a low predicted risk can never suppress urgent referral. The motivating vignette—resolved unilateral stress-associated tinnitus, then a delayed-care sudden loss in the other ear—is exactly the failure mode the red-flag layer exists to prevent. Public data identify population patterns;

the Ajou University Hospital extension can later test treatment timing, SSNHL recovery, speech-in-noise outcomes, rehabilitation, and tinnitus burden longitudinally.

7.4 Comparison with prior work

Unlike single-country prevalence studies (Park et al., 2014; Joo et al., 2015), StressEar-AI prespecifies cross-national transportability testing with calibration, recalibration, common support, and case-mix standardization—not discrimination alone. Unlike opaque clinical-AI demonstrations, it locks feature/outcome definitions before validation, declares model intended use (Table 5), publishes model cards, and treats subgroup fairness and safety evaluation as first-class results.

7.5 Limitations

This is a protocol, so its central limitation is that it reports *no empirical results*; its value rests on the rigor of the plan. Substantively: (1) the surveys lack validated job-strain instruments, so the exposure is perceived stress with work-related factors as covariates, and low measurement-comparability variables are excluded from cross-national pooling (Table 4); (2) cross-sectional audiometry cannot establish temporal order, so no causal claim is made despite the DAG-guided adjustment; (3) treating depressed mood/sleep as mediators rests on assumptions that may be violated, hence the separate confounder-treatment sensitivity analysis; (4) restricting to employed adults risks a healthy-worker/selection-collider effect, hence all-adult primary analysis; (5) survey-design and case-mix differences complicate transportability, addressed by the weighted/unweighted, common-support, and standardization analyses; (6) SHAP importance is descriptive and can be unstable under correlated predictors; (7) finite-set red-flag sensitivity of 1.0 does not guarantee zero deployment misses; and (8) open medical language models require local validation, privacy review, and governance before any use, and are confined here to clinician-verified research extraction.

7.6 Future directions (out of primary scope)

Beyond the three-axis core, future work may add longitudinal occupational cohorts to strengthen temporality; a neurophysiological substudy of tinnitus subtypes using open OpenNeuro/EEG datasets; audiogram-digitization to unlock legacy paper audiograms; and community extension of the harmonization schema to additional national surveys. These are deliberately kept out of the primary AJA analyses to preserve focus.

8 Conclusion

StressEar-AI is a pre-analysis protocol, not an empirical report: it fixes, in advance, a public-data-first and clinically grounded plan for studying associations among perceived stress/work-related factors, tinnitus, and hearing outcomes in adults. KNHANES is the development dataset and NHANES the external-validation dataset, with cycles, adjustment sets, endpoints, and metrics prespecified to guard against analyst degrees of freedom. Any open medical AI model accelerates clinician-verified feature extraction and explanation only, under schema constraints and a deterministic red-flag safety layer, and any prediction model is a research risk-stratification tool with declared intended use—never a diagnostic device. By registering aims, a causal DAG, survey-aware analysis, cross-national validation with recalibration and common-support checks, subgroup fairness, and a rigorously specified safety evaluation—and by releasing code, model cards, safety tests, and a synthetic-data path—the protocol maximizes transparency and reproducibility while avoiding unsupported causal claims. The planned public-data analyses will characterize associations and benchmark safe, explainable AI; the prospective Ajou University Hospital extension is required to study treatment timing and recovery—above all, to protect the narrow window in which sudden hearing loss is treatable.

Declarations

Author Contributions (CRediT): *Geunsik Lim*—Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing (original draft), Visualization. *Hyun Jo*—Clinical methodology (audiology/otolaryngology), Validation, Supervision of the planned clinical extension, Writing (review & editing). Both authors read and approved this draft.

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Funding: No external funding was received for this pre-analysis framework. Any funding for the prospective clinical extension will be disclosed at that stage.

Ethics Approval: Stages 1–2 use publicly available, fully deidentified survey microdata (KNHANES, NHANES) under each provider's data-use terms and do not constitute new human-subjects research. The Stage-3 clinical extension will obtain institutional review board approval at Ajou University Hospital before any data collection.

Reporting Guidelines: The observational analyses follow STROBE (von Elm et al., 2007); the prediction models follow TRIPOD+AI (Collins et al., 2024); the clinical extension will follow SPIRIT. A completed checklist is provided in the repository.

Protocol Status and Registration: This is a pre-analysis study protocol reporting no empirical results; all quantitative values are synthetic pipeline checks. The prespecified analysis plan (cycles, variables, adjustment sets, endpoints, and metrics) is time-stamped in the versioned repository, and public registration of the analysis plan is intended prior to first real-data analysis.

Data Availability: KNHANES data are available from the Korea Disease Control and Prevention Agency subject to official access procedures; NHANES public-use files are available from the U.S. CDC. The repository provides code, configuration, variable-mapping templates, and a synthetic-data generator, but does not redistribute restricted raw data.

Code Availability: All analysis code, configuration, model cards, prompt templates, and the reporting checklist are openly available at https://github.com/leemgs/

stress-ear-ai under an open-source license.

AI-Use Disclosure: Generative AI tools assisted in drafting and structuring this manuscript and the code scaffold; the authors reviewed and are responsible for all content. In the proposed study, open medical language models will be used only for research feature extraction and guideline-linked explanation from deidentified text, always with clinician verification before analysis and never for autonomous clinical decisions.

Conflicts of Interest: The authors declare no competing interests.

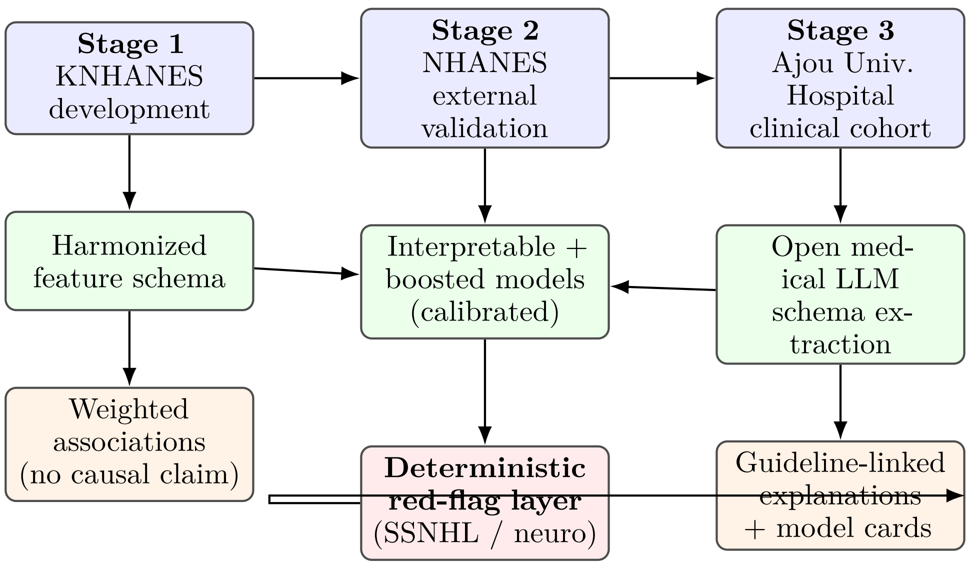

Figure 1: StressEar-AI conceptual framework. Public survey data (blue) feed a harmonized schema and calibrated, interpretable models (green); an open medical language model performs schema-constrained extraction in the clinical extension. A deterministic red-flag safety layer (red) can override probabilistic output to force urgent referral, so a low predicted risk never suppresses time-critical SSNHL care. Outputs (orange) are associations and guideline-linked explanations, not diagnoses.

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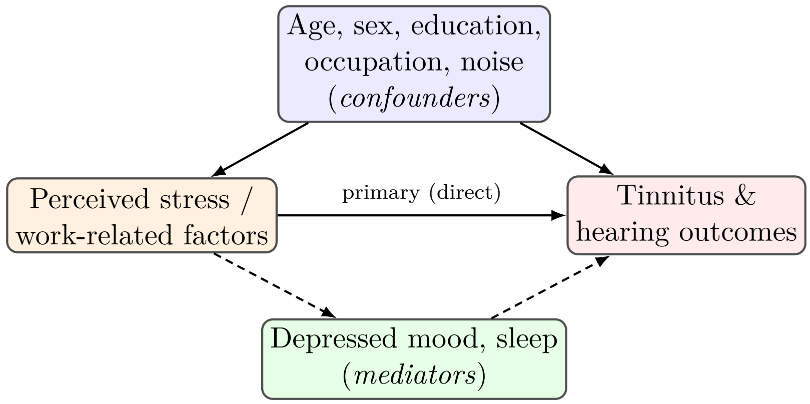

Figure 2: Assumed causal directed acyclic graph (DAG). The minimal sufficient adjustment set (blue confounders) is used in the *primary* analysis. Depressed mood and sleep (green) are treated as *mediators* of the stress→outcome path (dashed); conditioning on them (extended model) is a mediator-adjusted, potentially overadjusted estimate reported separately. Restricting to employed adults can open a selection/collider path, so all-adult analysis is primary and employed-restricted analysis is secondary.

Table 2: Selected public datasets for StressEar-AI

Dataset	Role	Strengths	Main limitations
KNHANES	Primary development dataset	Korean population relevance; audiometry in selected cycles; tinnitus and health variables; survey weights	Cross-sectional; stress variables may be perceived stress rather than validated occupational stress; raw data access procedures required
NHANES	External validation dataset	Public CDC files; audiometry and noise exposure in selected cycles; rich health covariates; geographic validation	U.S. population; cycle-specific hearing and tinnitus variables; complex survey design
OHHR / Zenodo audiology data	Secondary method testing	Audiology-specific measurements; useful for model benchmarking	May lack stress, occupation, or tinnitus variables
OpenNeuro tinnitus datasets	Optional neurophysiology substudy	Public neuroimaging or fNIRS data for tinnitus biomarkers	Not designed for occupational stress or longitudinal treatment outcomes

Table 3: Target harmonized variables, roles, and availability. Exposure and outcome roles follow the naming principle and the DAG (Figure 2): perceived stress is the exposure; work-related factors and the minimal confounder set are covariates; depressed mood and sleep are treated as mediators in the primary analysis. Availability is cycle-dependent; see Table 4 for source items and comparability.

Harmonized variable	Role	KNHANES	NHANES
Tinnitus presence	Outcome	Sel. cycles	Sel. cycles
Bothersome tinnitus	Outcome	Sel. cycles	Sel. cycles
Better-ear $PTA_{0.5,1,2,4}$	Outcome	Sel. cycles	Sel. cycles
Worse-ear PTA & per-ear thresholds	Outcome	Sel. cycles	Sel. cycles
Asymmetric hearing loss	Outcome	Sel. cycles	Sel. cycles
Self-reported hearing difficulty	Outcome (2^{nd})	Yes	Yes
Hearing-aid use	Outcome (2^{nd})	Sel. cycles	Sel. cycles
Perceived stress	Exposure	Yes	Yes
Work-related factors (occupation, employment, hours)	Covariate / modifier	Yes	Yes
Occupational + recreational noise (duration/intensity)	Confounder	Sel. cycles	Yes
Depressed mood	Mediator	Yes	Yes
Sleep duration	Mediator	Sel. cycles	Sel. cycles
Smoking, alcohol, cardiometabolic disease	Confounder	Yes	Yes
Age, sex, education	Confounder	Yes	Yes
Socioeconomic status; race/ethnicity	Fairness axis	Partial	Yes
Survey weight, stratum, PSU	Design	Yes	Yes

Table 4: Cross-national harmonization and measurement-comparability plan (excerpt). Exact cycle-specific variable codes and response categories are pinned in the repository data dictionary. Comparability governs use: *low*-comparability variables are excluded from the common transportable model and entered only in country-specific extended models. "Sel. cycles" = present in selected cycles only.

Harmonized variable	KNHANES (source/cycle)	NHANES (source/cycle)	Original item (abridged)	Transformation rule	Compar.
Tinnitus presence	Otologic Q, 2009–2012	Audiometry Q (AUQ), sel. cycles	"Ringing/noise in ears past 12 mo?"	Any vs. none (binary)	High
Bothersome tinnitus	Otologic Q, 2009–2012	AUQ bother item, sel. cycles	Annoyance / sleep / concentration	Bothersome vs. not	Medium
Pure-tone thresholds	Audiometry exam, 2010–2012	Audiometry exam, sel. cycles	Air-conduction dB HL per freq.	Better/worse-ear PTA; per-ear	High
Perceived stress	Health Q (stress item)	DPQ / stress-related item	"How much stress in daily life?"	Collapse to high vs. low	**Low**
Depressed mood	PHQ-9 / mood item	PHQ-9 (DPQ)	Standardized depression items	Screen-positive vs. negative	Medium
Occupation / employment	Socioecon. Q	Occupation (OCQ)	Job category, employed y/n	Common category crosswalk	Medium
Occupational noise	Sel. cycles	Noise exposure (AUQ)	Loud-noise exposure, duration	Duration/intensity + binary	Medium
Sleep duration	Health Q	Sleep (SLQ)	Average hours of sleep	Continuous hours	Medium
Smoking / alcohol	Health Q	SMQ / ALQ	Current status, quantity	Standard categories	High
Age, sex, education	Demographics	Demographics (DEMO)	—	Direct / crosswalk	High
Survey weight, stratum, PSU	Design vars	Design vars	—	Kept for design-consistent SEs	High

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Table 5: Intended-use specification for the two prediction models. Both are *research risk-stratification* tools only; neither is a screening, diagnostic, or triage device. The specification is fixed before validation so that performance is judged against a declared purpose.

Dimension	Specification
Purpose	Research risk stratification for cohort description and hypothesis generation; *not* individual screening, diagnosis, triage, or treatment guidance.
Target user	Researchers/epidemiologists analyzing survey or, later, deidentified cohort data.
Prediction time	Cross-sectional (concurrent survey visit); no temporal/prognostic claim.
Target population	Adults 40–69 y in the fixed audiometry windows; not generalized beyond.
Inputs	Prespecified harmonized covariates (Table 3); no free-text or image inputs in the public-data models.
Output / decision	Calibrated probability for research stratification; drives *no* clinical action.
Cost of false positive	Low in research use; would be non-trivial if misused for screening (over-referral) — such use is contraindicated.
Cost of false negative	Low in research use; dangerous if misused clinically (missed disease) — such use is contraindicated.
Contraindicated uses	Individual diagnosis, treatment decisions, insurance/employment decisions, or any deployment without prospective clinical validation and regulatory review.
Safety interaction	The deterministic red-flag layer (Table 8) overrides model output for time-critical symptoms regardless of predicted probability.

Table 6: AI components and safety constraints

Component	Intended use	Safety constraint
Survey harmonization	Map KNHANES/NHANES variables to common schema	Human-readable mapping file and version control
Statistical baseline	Estimate adjusted associations and transparent risk scores	Survey-aware methods; no causal claims from cross-sectional data
MedGemma or similar open medical LLM	Extract fixed fields from deidentified clinical notes in future cohorts	Clinician verification; no diagnosis or treatment recommendation
Retrieval-augmented explanation	Draft evidence-linked explanations for researchers and clinicians	Curated sources only; unsupported outputs flagged
Red-flag rule layer	Identify sudden hearing-loss or neurologic warning signs	Deterministic referral rules override probabilistic model

Table 7: Prespecified evaluation metrics (frozen before external validation). The right column shows *synthetic-scaffold* values that verify pipeline wiring only: they are computed on randomly generated data, carry **no** epidemiological meaning, and must not be read as results. Real-data estimates replace them when KNHANES/NHANES files are supplied.

Dimension	Metric(s)	*Synthetic scaffold (not a result)*
Discrimination	AUROC, AUPRC (unweighted and survey-weighted)	Tinnitus: 0.64 / 0.49
Calibration	Calibration-in-the-large and calibration slope, reported separately, before and after recalibration; Brier score; plots	Tinnitus Brier: 0.20
Transportability	Common-support region; case-mix-standardized performance; reduced-common vs. country-specific model	Reported per target
Clinical utility	Decision-curve net benefit	Reported per target
Interpretability	SHAP importance (descriptive; caveated for correlated predictors)	Reported per target
Fairness	Subgroup AUROC & calibration: sex, age, employment, noise, asymmetry, SES, race/ethnicity	Reported per subgroup
LLM extraction	Exact match; precision/recall/F1; omission; span agreement; temporal/laterality/onset/duration accuracy; doc-level match; run consistency; clinically significant error rate	Clinical extension
Safety layer	Sensitivity (95% CI); specificity/over-referral; subgroup & robustness (Table 8)	Target $\rightarrow$ 1.0

Table 8: Prespecified evaluation plan for the deterministic red-flag (SSNHL/neurologic) safety layer. Sensitivity (red-flag recall) is the primary metric because a missed red flag is the costly error; but a perfect score on a finite test set does not guarantee zero real-world misses, and this limitation is stated explicitly.

Evaluation element	Prespecification
Vignette set	≥200 vignettes; composition fixed: real deidentified clinical cases, expert-authored cases, and synthetic cases in a stated ratio (e.g., 40/30/30%).
Reference labeling	Each vignette independently labeled by ≥2 otolaryngology/audiology experts; disagreements adjudicated.
Inter-rater agreement	Reported (Cohen's/Fleiss' κ) before adjudication.
Primary metric	Sensitivity (red-flag recall) with 95% CI.
Secondary metrics	Specificity and over-referral rate with 95% CIs.
Subgroup performance	By laterality (unilateral/bilateral), onset (acute/gradual), and vertigo/neurologic signs.
Robustness	Negation, ambiguous temporal expressions, and colloquial patient phrasing explicitly tested.
Error analysis	Every missed red flag audited and reported as a critical failure.
Stated limitation	Finite-set sensitivity of 1.0 is a target, not a guarantee of zero misses in deployment.

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